

Full Functionality Guide — Every Page, Flow & Scenario

Product: **Que** (SchemaGraph) — Cursor for data teams · schema-first · HITL · attested delivery

Local app: <http://localhost:5174> · API: <http://localhost:8787>

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Audience: DE / DA / analytics eng / stewards / admins / pilot leads

Purpose of this document: teach every production surface so a team can *perform daily work* and *upskill* — not a marketing brochure.

Honest boundaries: Que is not your warehouse, not Tableau, not SOC 2 Type II by itself. AI never reads the full lake — only metadata + pinned scrubbed samples (5–10 rows).

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1. Quick start & demo login

Piece	Command / URL
Postgres	Docker <code>stitch-pg</code> on <code>:5432</code> (user/db/pass: <code>stitch</code>)
Migrate	<code>cd api && npm run migrate</code>
Seed demo owner	<code>npm run seed:demo</code> → <code>demo@que.local</code> / <code>que-demo-2026</code>
API	<code>cd api && npm run dev</code> → <code>http://localhost:8787</code>
UI	<code>npm run dev</code> → <code>http://localhost:5174</code>
Health	<code>GET /health</code> · <code>GET /status</code> · public UI <code>/status</code>

Legacy demo users may also exist (e.g. `owner@stitch.local`). Prefer the Que demo seed for client walkthroughs.

2. Product DNA & Offer A / B

Rule	Why it matters in production
Schema-first	Connectors introspect structure; they do not ETL your lake into Que.
Pinned samples 5–10	Scrubbed, frozen until re-pin; used for join overlap ~88–95% band.
Human Promote	Suggested joins are not truth until Promote (HITL).
Attested delivery	Exports / jobs carry evidence; warehouse (or managed plane) is SoR for rows.
AI denied lake	<code>aiMayUsePinnedSamples</code> default ON; managed row payloads denied to AI.

Offer A — Bring your warehouse

Databricks / Snowflake / Postgres stay system of record. Que proposes joins, drafts jobs, digests external run failures, exports / triggers.

When: You already have a warehouse and want control-plane HITL.

Offer B — Que managed plane

Excel/SQL teams land certified datasets in Que (quotas + retention). Certify → BI charts → embed.

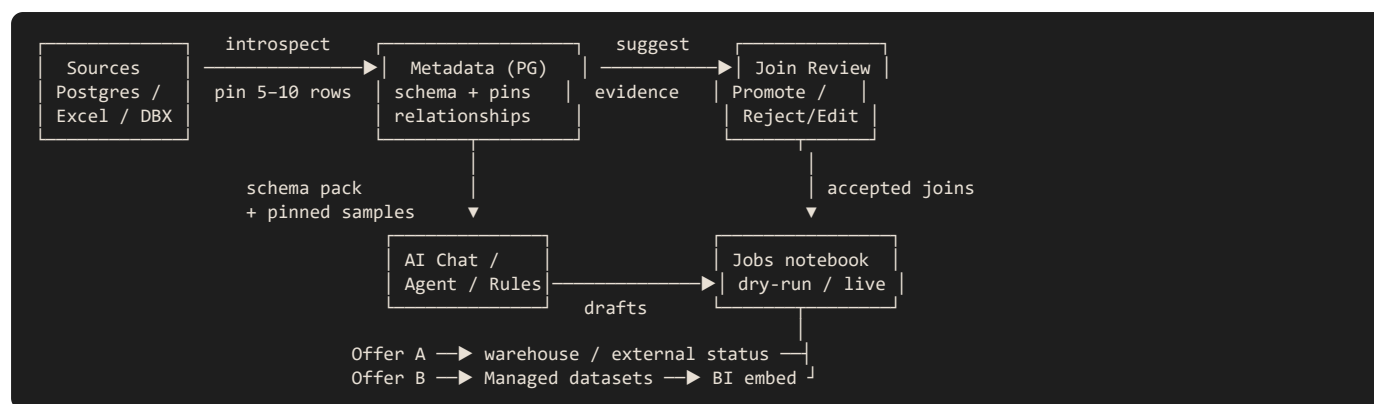
When: No warehouse yet, but need attested DA path.

3. Roles & permissions

Role	Can do	Cannot do
Viewer	Browse schema, joins, lineage, BI preview (read)	Sync, Promote, settings writes, mint embeds
Member	Sync, edit joins, Promote/Reject, jobs, transforms, comments	Most admin security / billing / retry policy
Admin	Members, security, enterprise, DR/backup, embed tokens, golden schedule	Owner-only destructive workspace ops (as configured)
Owner	Full workspace control + billing	—

Presence chips in the top bar show teammates currently active (HTTP heartbeat — not live cursors).

4. Architecture & master flows



```
CURSOR LOOP (team OS)
Rules (memory) → Chat/Agent drafts → Transforms / Proposals (diff) → Approve
^
+----- learn-from-Promote <-----+
Eval (golden schedule) monitors join quality over time
```

5. Scenario A — Offer A warehouse stitch (production day)

Goal: Two sources → trusted joins → job → attested handoff into customer warehouse; failures digested in Que.

- 1 Login as owner/admin → open `/product` once for positioning, then `/sources`.
- 2 Add Postgres + Excel (or Databricks/Snowflake). Click **Sync schema**. Confirm pins created (scrubbed samples).
- 3 Go to `/joins`. Filter **Suggested**. Open a high-score edge. Review evidence + overlap.
- 4 If columns wrong: edit mapping → save → **Promote**. Discuss in thread comments with DE/DA.
- 5 `/settings/ai-policy`: set execution plane to **customer** (Offer A). Keep pinned samples ON.
- 6 `/jobs`: create or open job from stitch / marketplace pack. Dry-run (samples). Live validate only when ready.
- 7 Export / trigger warehouse job. Post status via external-status bridge (or automation).
- 8 `/compliance` → **Build digest** for Offer A failure summary. Chat shows Offer A strip too.
- 9 `/lineage` + export attestation verify at `/verify` for auditors/partners.

Success: Promoted joins feed job contract; warehouse remains SoR; Que holds decisions + evidence.

6. Scenario B — Offer B managed plane + certified BI

Goal: Excel/SQL team lands a certified table in Que and embeds a chart for stakeholders.

- 1 Enable managed plane in `/settings/ai-policy` (`enableManagedDataPlane`).
- 2 Connect Excel/CSV → sync → promote joins → job dry-run → **Land to managed** (from job flow).
- 3 `/managed`: open dataset → review columns → **Certify**.
- 4 `/metrics`: define `COUNT(*)` or `SUM(amount)`, optional source column for lineage → Certify → Publish BI.
- 5 `/bi`: bind certified dataset, set chart type/axes, **Preview**, **Certify**, **Mint embed**.

6 Share viewer URL `/embed/<token>` (public, read-only). API also at `GET /bi/embed/:token`.

7 Watch quotas/retention on managed plane; purge expired from ops/compliance path as needed.

AI still cannot pull managed row payloads. Embeds are for humans/stakeholders.

7. Scenario C — Team Cursor loop (daily DE/DA)

1 Admin seeds org `/rules` (join conventions, naming, PII).

2 DE opens `/agent` or `/chat` with `@tables` → draft transform `/job`.

3 Transform lands in `/transforms` and/or `/proposals` as open diff.

4 Peer reviews unified diff on `/proposals` → Approve / Reject.

5 On Promote, rules can learn patterns. Comments/threads on `/joins` capture debate.

6 `/eval`: configure golden pairs JSON → enable schedule → Run now; watch recall.

7 `/marketplace`: install industry pack to bootstrap a job when starting a domain.

8. Every page — detailed production use

8.0 Public pages (no login)

`/login` — **Login**

What: Email/password and optional SSO continue.

Do: Sign in → lands on Workspace. Use SSO when org enforces it.

Scenario: New hire invited → accepts invite → login → switches workspace if multi-tenant.

`/sales` — **Public sales**

Honest positioning for prospects (Offer A/B). CTA to login / status. Not a full marketing site.

`/status` — **Public status**

Ops transparency: API ok, DB latency, inventory counts, region. Share with clients during incidents.

`/verify` — **Attestation verify**

Paste/export fingerprint or token to verify attested artifacts independently of login.

`/embed/:token` — **Certified BI viewer**

Read-only chart for stakeholders. Token minted only after chart certify. Revoke from BI admin actions.

8.1 `/workspace` — Schema canvas

Purpose

Visual system of record for *schema truth*: tables as nodes, relationships as edges.

What to do

- Pan/zoom canvas; drag tables (members+); expand columns.
- Select table → right inspector: schema, samples (capped), add to job.
- Filter by source; use TopBar confidence / join filters when available.
- Viewers: read-only (no drag persist).

Production scenario

After sync, DE opens Workspace to sanity-check that Excel + Postgres tables both appear, then jumps to Joins for HITL.

8.2 /sources — Connectors & sync

Purpose

Register connectors and sync **schema + pinned samples**.

Connectors (typical)

PostgreSQL, Excel/CSV, MongoDB, Databricks, Snowflake, BigQuery, Salesforce (fixture and/or live).

How to use

1. **Add source** → pick connector → fill form (or fixtures path for demos).
2. **Sync schema** — retries/checkpoints apply automatically.
3. Set sync schedule (hourly/daily/off) for continuous introspect.
4. On error: check health kind (auth vs transient); fix secrets; re-sync.

Scenario

Ops schedules nightly sync for Snowflake. Auth failure stops tight loops; transient errors retry with backoff.

SLA view: API /connector-reliability also surfaces on Compliance.

8.3 /joins — Join Review (HITL)

Purpose

The trust gate. Suggested joins become production truth only after Promote.

Controls

- Filters: suggested / accepted / rejected / all
- Evidence: name/type, uniqueness proxy, pinned overlap score
- Edit columns / direction → save → Promote or Reject
- Team discussion: comment + **Reply** threads
- Re-run join inference / mapping assist / golden eval hooks as available

Scenario

DA believes `customer_id` ↔ `cust_id`. Overlap 91%. DE edits left/right columns, threads “confirmed in CRM”, Promotes. Proposal/diff may also appear for audit.

Never claim 100% join accuracy. Band ~88–95% on easy keys with pins.

8.4 /chat — AI Chat

Purpose

Schema-only assistant: design questions, SQL drafts, job drafts. Uses schema packs + optional pinned samples.

How to use

- @mention tables; slash skills; sidebar focus tables
- Ask for join rationale, notebook steps, validation checks
- Save job draft when offered (members+)
- Offer A strip: refresh warehouse digest without leaving chat

Scenario

“Draft a reconciliation job using promoted joins between ledger and bank_feed” → review SQL → save job → open Jobs.

8.5 /agent — Stitch Agent

Purpose

Multi-step agent sessions (checkpoints) for stitch goals, including `draft_transform`.

How to use

1. Create session with a clear goal (“transform events to weekly active accounts”).
2. Advance checkpoints; when joins needed, Promote in Joins then continue.
3. Resulting transform/proposal appears in Transforms / Proposals for HITL.

8.6 /jobs — Notebook → ship

Purpose

Production stitch jobs: markdown + SQL notebook, freeze contract, dry-run, live validate, export, schedule, materialize.

Tabs / actions (typical)

- Notebook editor (SQL cells)
- Dry-run (capped samples) vs live (bounded rows)
- Export JSON / SQL / dbt / dbt PR (gates: promoted joins, drift)
- Schedules + private runner / orchestrator hooks
- Templates panel; land to managed when Offer B
- Contract tests run history

Scenario

After Promote, open job → dry-run → fix null keys → export dbt PR → warehouse CI runs → external status posts back → digest on Compliance.

8.7 /proposals — Approve / Diff

Purpose

PR-style inbox for joins, transforms, SQL changes.

How to use

- Toggle **Unified** vs side-by-side
- Approve / Reject (members+)
- Link to Agent for drafting context

Scenario

Junior DA drafts transform; senior DE reviews unified diff; Approve applies workspace truth.

8.8 /transforms — NL → SQL HITL

Purpose

Natural-language transform drafts with review states (proposed → approved/rejected/applied).

Scenario

“Weekly active accounts from events” → draft SQL → review → approve → attach/open job.

8.9 /rules — Org memory

Purpose

Cursor-like rules injected into AI + team conventions. Can learn from Promote.

How to use

- Create join / naming / PII rules; enable/disable
- After Promote, confirm learned rule suggestions

Scenario

Rule: "Never join on email alone for finance." Chat/agent respects it in drafts.

8.10 /metrics — Semantic layer

Purpose

Certified metrics on managed datasets; lineage-to-metric graph.

How to use

1. Create metric name + expression (`COUNT(*)` , `SUM(amount)`)
2. Bind certified managed dataset; optional source column
3. Preview → Certify → Publish BI
4. Inspect metric lineage edges on the page

8.11 /bi — Certified BI

Purpose

Lightweight charts (table/bar/kpi/pie) on certified managed data; embed tokens.

How to use

1. Create chart → bind certified dataset → set type/x/y
2. Preview → Certify → Mint embed (admin)
3. Copy viewer URL `/embed/...` and API GET URL

Not a Tableau replacement — stakeholder embeds with certification gate.

8.12 /managed — Offer B datasets

Purpose

Host job outputs with quotas/retention when no warehouse SoR.

How to use

- List datasets; open rows (capped UI); certify; delete when obsolete
- Only certified datasets feed BI / metrics publish

8.13 /marketplace — Industry packs

Purpose

Install starter job packs (retail, finance, SaaS, healthcare, logistics, marketing, HR, platform, Snowflake/dbt, Databricks gold).

How to use

- Filter industry/tag/search → Install → opens draft job
- Review recent installs list

Scenario

New finance domain: install Ledger reconciliation → sync real tables → remap joins → Promote → run.

8.14 /eval — Quality dashboard

Purpose

Promote rates, job success, rules coverage + scheduled golden-set eval.

How to use

- Read KPI cards weekly in ops review
- Admin: paste golden pairs JSON → Save & enable → Run now
- Recall drops are audited (alerts path)
- Quick-apply industry templates (or use Marketplace)

8.15 /lineage — Lineage lite

Purpose

Metadata lineage (not warehouse-native full column graph for every vendor).

Scenario

Before export, DA checks which tables feed the job; pairs with column-lineage API when investigating a metric.

8.16 Catalog · Glossary · Domains · Steward

Page	Use in production
/catalog	Asset inventory / governance hooks — describe products, not replace Workspace stitch.
/glossary	Business terms aligned to tables/columns for DA clarity.
/domains	Data-product domains; ownership boundaries for multi-team workspaces.
/steward	Stewardship queue: ownership, quality follow-ups, policy actions.

8.17 /validation & /drift-agent

Validation suite

Run structured checks before ship (nulls, dups, contract expectations).

Drift agent

When source schema changes after sync, review drift severity; alerts can webhook/email (settings).

Scenario

Upstream renamed column → sync drift high → block export → fix joins → re-Promote → re-export.

8.18 /compliance — Evidence & ops checklist

Purpose

Auditor-facing evidence pack + SaaS ops checklist + Offer A digests. **Not** a Type II letter.

How to use

- **Refresh pack** / Download markdown
- Admin: **Run backup**, **Run DR drill**
- **Build digest** for warehouse/external failures
- Read connector SLA summary

8.19 /product — Client positioning

Internal/client conversation page: Offer A/B, day-one path, what you do not claim. Use at kickoff meetings.

8.20 Settings — every tab

Route	What to configure	Who
/settings/members	Invite users, roles, revoke	Admin+
/settings/security	SSO enforce, sessions, secrets	Admin+
/settings/enterprise	CMK, SCIM, SIEM, ABAC, isolation tests, SOC2 export	Admin+
/settings/automation	Sync schedules overview, job schedules, orchestrator, private runner	Admin/Member
/settings/governance	Policy packs, PII, tickets, certifications	Admin+
/settings/team	Team OS notify (Slack/Teams), promote min role	Admin+
/settings/billing	Seats / portal / checkout	Owner/Admin
/settings/ai-policy	Pinned samples, scrub, managed plane, auto-promote (keep OFF), LLM prefs	Admin+

Keep **auto-promote low-risk** off unless you deliberately accept risk and document it.

9. Ops, health, digests, DR

Capability	Where	Production habit
Health / status	/health , /status , UI /status	Uptime checks + status page link in runbooks
Prometheus	/metrics?format=prom	Scrape from your observability stack
Warehouse digests	Compliance + Chat strip + ingest API	Daily for Offer A heavy warehouses
Metadata backup	Compliance → Run backup	Weekly + before major schema campaigns
DR drill	Compliance → Run DR drill	Quarterly (records evidence)
Golden eval	Eval schedule	Daily/weekly; investigate recall drops
Managed retention	API purge / settings days	Align with DPA retention

10. Team operating model & upskilling

WEEKLY RITUAL (recommended)
Mon Sync health + digests review (Ops/DE)
Tue Join Review queue burn-down (DE + DA)
Wed Proposals/Transforms review hour (senior DE)
Thu Metrics/BI certify for stakeholders (DA)
Fri Eval recall + Rules cleanup + Compliance checklist

Role upskill path	Practice in Que
Junior DA	Marketplace pack → Chat drafts → comment on joins → never Promote alone first week
DE	Edit/Promote joins → Jobs dry-run → dbt export → digest failures
Analytics eng	Metrics + BI certify + embeds; lineage checks
Steward / governance	Glossary/domains/steward + Compliance pack
Admin	SSO/SCIM/SIEM, DR drills, golden schedule, AI policy

Upskill metric: approvals/week, promote rate, job success 30d, golden recall trend, time-to-first-certified-BI.

11. Troubleshooting cheat sheet

Symptom	Likely cause	What to do
Sync auth error	Bad token/password	Fix secrets; auth errors skip tight scheduled loops
Low join confidence	Weak keys / empty pins	Re-pin samples; edit mapping; use query-history assist on DBX/SF
Export blocked	Unreviewed joins / drift	Promote or reject; resolve drift; check settings gates
BI mint fails	Chart/dataset not certified	Certify dataset → chart → mint
AI refuses rows	By design	Use pins only; never expect lake dump
Viewer can't Promote	Role	Ask admin for member+
Embed 401	Revoked/expired token	Mint new token; check certify still true
Golden eval 400	No pairs	Paste pairs JSON on Eval before Run

12. Claims you can / cannot make

You can say • Schema-first HITL stitch control plane • Pinned scrubbed samples; ~88–95% band on easy keys • Attested exports into customer SoR • Evidence pack for auditor diligence • Offer A + Offer B dual path	Do not say • We replace Databricks/Snowflake • AI is always 100% correct / no humans • We are SOC 2 Type II certified (unless auditor letter exists) • Que is the enterprise lake for all rows • Certified BI is a full Tableau suite
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13. Deep control inventory (button / field level)

Use this section as a training checklist — walk each control once on a demo workspace.

13.1 App chrome (every authenticated page)

Control	Action
Que logo	Go to Workspace
Workspace switcher	Switch membership context (multi-workspace users)
Primary nav links	Jump to Workspace, Chat, Sources, Joins, Proposals, Transforms, Rules, Metrics, Eval, Marketplace, Domains, Catalog, Jobs, Managed, BI, Compliance, Product, Lineage, Steward, Agent, Settings
Presence chips	See who is active + which path they are on
Onboarding roadmap	Open guided checklist dialog
Auth session controls	Account / logout / session management
Mobile hamburger	Same nav on narrow viewports

13.2 Workspace canvas

Control	How to use
Background drag	Pan diagram
Wheel / zoom buttons	Zoom in/out
Drag table header	Reposition node (saved for member+)
Click table title	Select + expand/collapse columns
Click column	Inspect column detail / samples
Source filter	Focus one connector's tables
Add to Job	Push selected tables into a job draft
Show Lineage	Jump toward lineage investigation

Training drill

- 1 Sync two sources → open Workspace → confirm both source colors/icons.
- 2 Expand a fact + dim table → note key icons.
- 3 Select relationship edge if shown → jump to Joins for Promote.

13.3 Sources

Control	How to use
Add / New source	Pick connector type
Connector form fields	Host, token, warehouse, fixtures path, etc.
Sync schema	Introspect + pin samples (retries automatic)
Sync schedule	off / hourly / daily
Status LED	active / warning / error
Delete connection	Admin; removes metadata for that source

Training drill — Excel + Postgres

- 1 Add Postgres demo fixture → Sync → note table count.
- 2 Add Excel/CSV pack → Sync → pins created.
- 3 Set Postgres schedule to daily → confirm next run time on Automation settings.

13.4 Joins

Control	How to use
Status filter chips	suggested / accepted / rejected / all
Queue list	Select an edge to inspect
Evidence panel	Read score, signals, overlap
Edit columns	Change left/right column mapping
Promote	Accept into workspace truth (member+ / min role)
Reject	Store negative memory for inference
Comment / Reply	Threaded discussion before Promote
Re-run inference	Refresh suggestions after new sync
Mapping assist	Rename / alias suggestions review

13.5 Chat

Control	How to use
Composer	Natural language; @table mentions
Skills	Slash/skill shortcuts for common asks
Sidebar tables	Focus context pack
Save job draft	Materialize assistant draft into Jobs
Reindex	Rebuild embeddings after big sync
Stop	Cancel in-flight ask
Offer A strip	Refresh warehouse digest
Feedback	Thumbs / notes for model quality

13.6 Jobs

Control	How to use
Job list / open	Select notebook job
Notebook cells	Edit SQL / markdown
Dry-run	Bounded sample execution
Live validate	Bounded live rows (careful)
Export JSON/SQL/dbt/PR	Attested handoff (gates apply)
Schedule	Cron-like job schedule
Templates	Insert common notebook patterns
Land managed	Offer B materialize into Que plane
Contract tests	Run + view history
External status	Bridge warehouse run state into Que

13.7 Marketplace → Eval → Metrics → BI chain

```
Install pack (/marketplace)
-> draft job (/jobs)
-> map real tables + Promote (/joins)
-> dry-run
-> (Offer B) land + certify (/managed)
-> define metric (/metrics) + lineage
-> chart + certify + mint (/bi)
-> share /embed/:token
-> track quality (/eval golden schedule)
```

13.8 Compliance ops drill (quarterly)

- 1 Refresh SOC2 evidence pack → download markdown → attach to diligence folder.
- 2 Run metadata backup → confirm checklist “backup” done.
- 3 Run DR drill → confirm passed steps (backup + isolation + db ping).
- 4 Build warehouse digest → file Offer A failures with owners.
- 5 Confirm connector SLA counts (ok/degraded/breached).
- 6 Note: this is evidence, not Type II certification.

14. Glossary of Que terms

Term	Meaning
Pin / pinned samples	Frozen scrubbed 5–10 rows per table for overlap/AI
Promote	Human accept of a suggested join into truth
Contract freeze	Snapshot of joins/schema bound into a job
Attestation	Signed evidence on exports for verify
Offer A	Customer warehouse is SoR
Offer B	Que managed plane hosts certified datasets
Proposal diff	PR-like before/after for review
Golden eval	Labeled join pairs to measure recall
Digest	Aggregated external/warehouse run failures
Presence	Heartbeat who is online in workspace

15. API quick reference (for power users)

Method	Path	Use
GET	/status / /health	Ops health
POST	/workspaces/:id/connections/:id/sync	Schema sync w/ retries
POST	.../relationships/:id/review	Promote/reject
GET/POST	.../comments	Join threads
GET	/marketplace/packs	Catalog
POST	.../industry-templates/:packId/apply	Install pack
GET	.../metrics-lineage	Semantic lineage
POST	.../presence/heartbeat	Multiplayer presence
POST	.../warehouse-digests/build	Offer A digest
POST	.../dr-drills/run	DR evidence
GET	/bi/embed/:token	Public embed JSON

How to train a new teammate in one day

- Morning: Product page + Scenario A through Promote (sections 2, 5, 8.2–8.3).
- Midday: Chat → Jobs dry-run → export (8.4, 8.6).
- Afternoon: Marketplace pack OR Offer B BI path (6, 8.11–8.13).
- End of day: Eval glance + Compliance backup (8.14, 8.18) + read claims (12).